EAST WEST UNIVERSITY

Department of Computer Science and Engineering

Architecture Pattern

Course Name: Software Architecture
Course Code: CSE-423
Section No: 01

Architectural pattern lab tasks to do:
1. Architectural context diagram, defining the archetypes, breakdown into components (3 diagrams) on the project you are working on.
2. Design the system architecture of the project you are working on with one or many suitable architectural diagram(s).

Group-4

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1.

1.1 Architectural Context Diagram

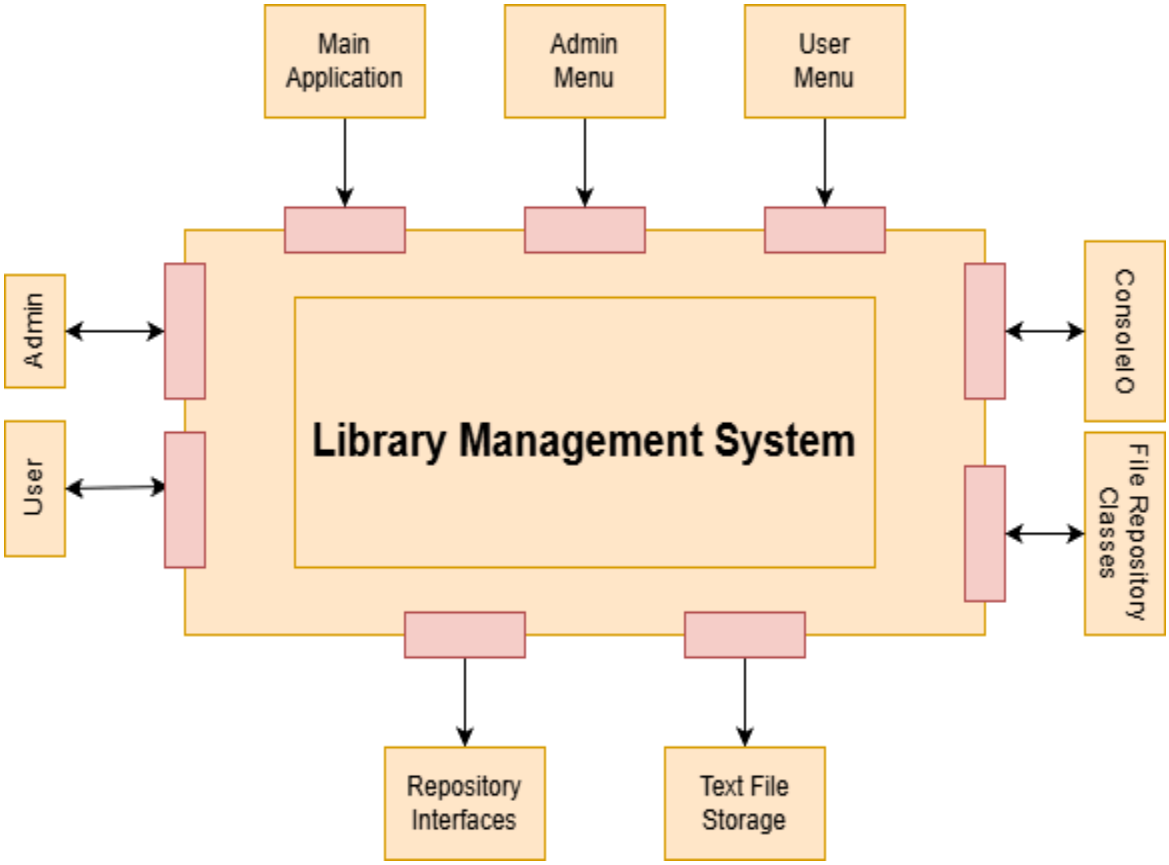

Fig-01 Architectural Context Diagram

1.2 Archetypes

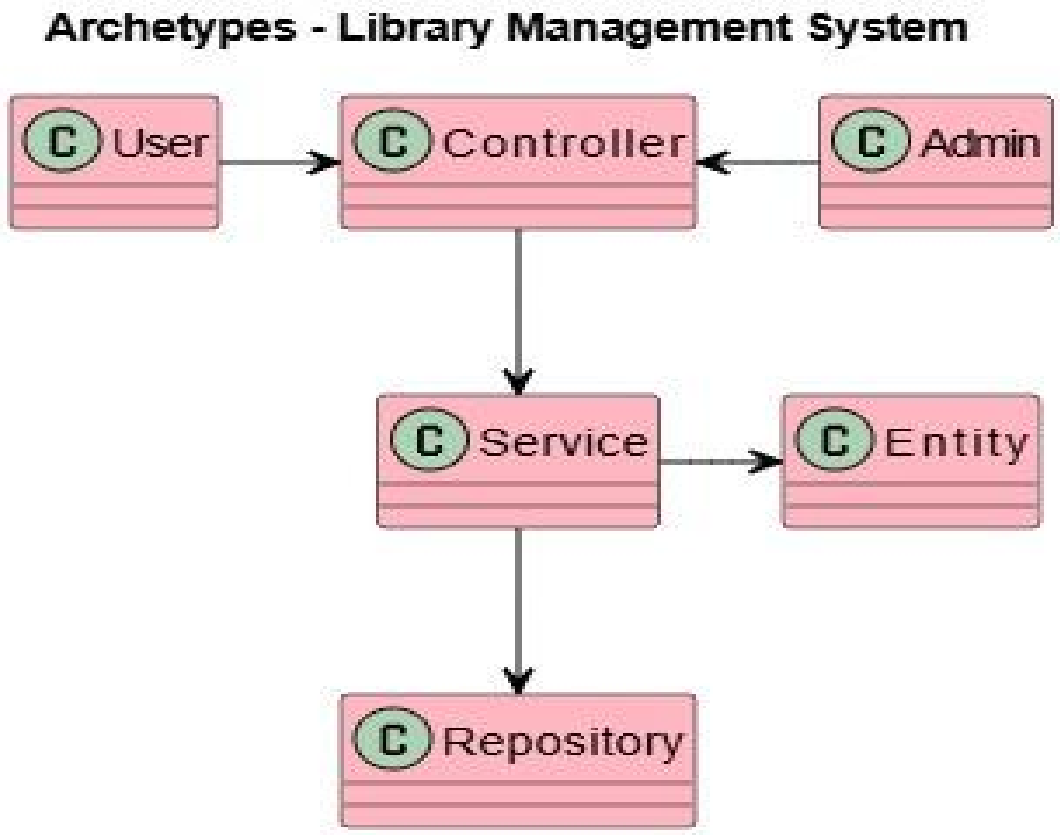

Fig-02 Archetypes

1.3 Architecture Into Components

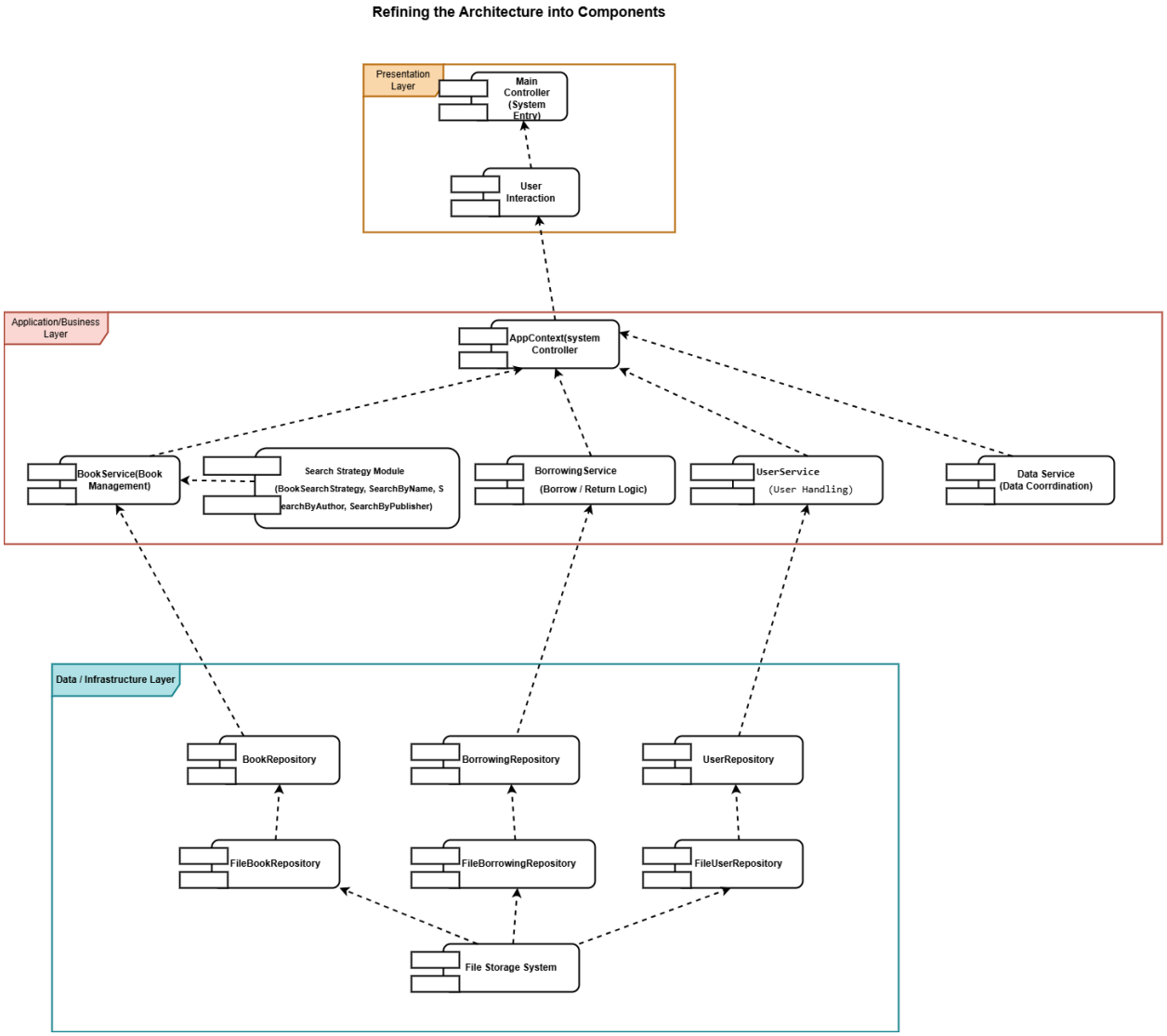

Fig-03 Architecture Into Components

2.

For this Library Management System the best architectural pattern is a Layered Architecture Pattern.

Because this project is already divided into clear layers:

1. **UI / Operation Layer**
   Handles user actions like add book, borrow book, place order, view orders.
2. **Service Layer**
   Contains business logic like user service, book service, borrowing service, order service, data service.
3. **Repository Layer**
   Handles data access using file-based repositories.
4. **Data Storage Layer**
   Stores data in files.

Layered Architecture-Library Management System

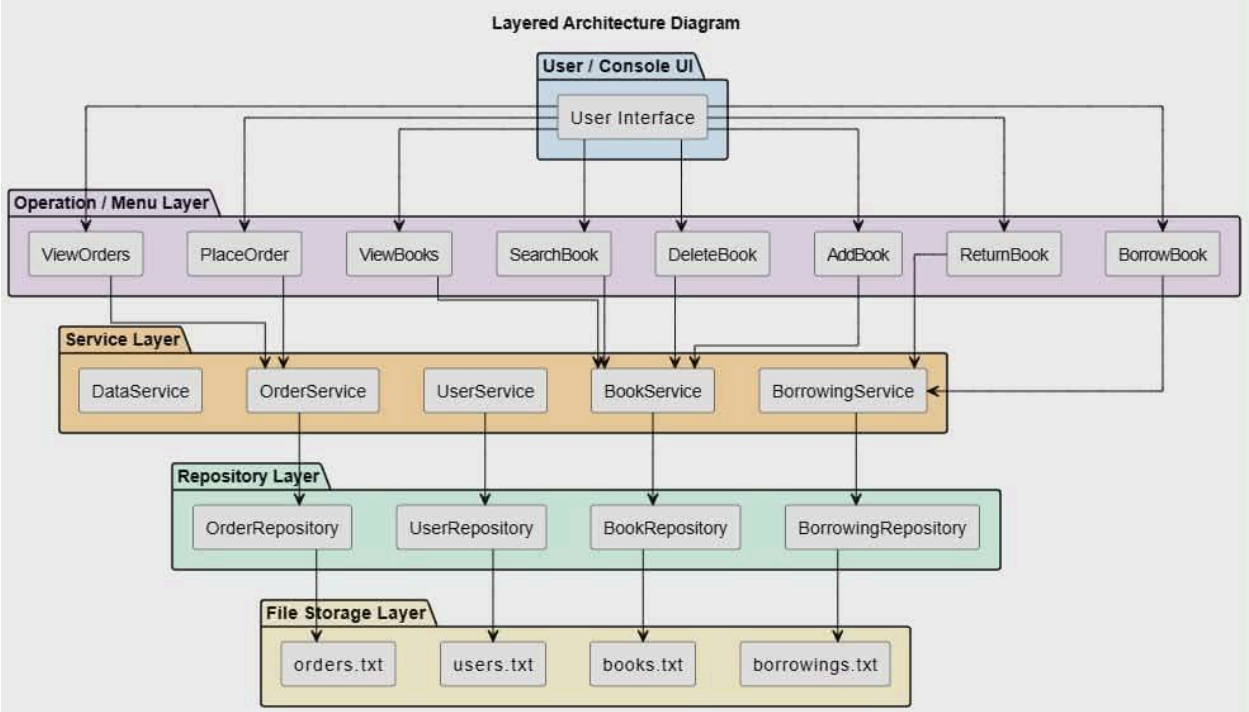

Fig-04 Layered Architecture Pattern